

11. Build the Coil

The machine everything else was for

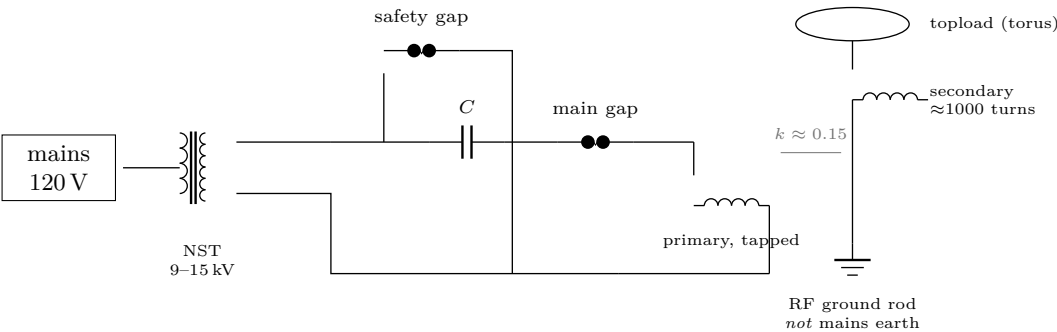
A small spark-gap Tesla coil, and how to tune it

Lesson 11 · several weekends · adult owns the mains side, permanently

ADULT-OPERATED: MAINS-DERIVED HIGH VOLTAGE

From here on the apparatus contains voltages and currents that kill people every year. The adult owns the mains plug, the transformer and every decision about energizing. Nobody touches anything inside the enclosure until the supply is unplugged *and* the capacitor has been shorted with the grounding stick. Read the **Safety** sheet completely before this lesson, not during it.

The circuit



Parts

Part	Typical for a small coil	Notes
Transformer	9 kV, 30 mA neon-sign transformer	An older non-GFCI unit. A modern electronic sign transformer will not survive this duty. This is the lethal component.
Tank capacitor	5–10 nF at 20 kV+, as a series-parallel string of pulse-rated film capacitors (an “MMC”)	Never electrolytic, never ordinary film. Each capacitor gets its own 10 MΩ bleeder resistor across it.
Main gap	Two 1–1.5 in copper pipe electrodes, adjustable, or a multi-gap string	Round and smooth, mounted so the spacing can be set repeatably. Fan-cooled if you run it more than briefly.
Safety gap	Fixed, set slightly wider than the main gap	Protects the transformer from reflected spikes. Not optional.
Primary	10–15 turns of 1/4 in copper tube or heavy wire, flat spiral, <i>tapped</i>	The tap is the tuning control. Leave it accessible.
Secondary	900–1200 turns of 24–26 AWG enamelled wire on 3–4 in PVC or acrylic, aspect ratio about 4:1 or 5:1	Close-wound, no gaps, no overlaps. Sealed with polyurethane afterwards.
Topload	Aluminium dryer duct formed into a torus, or a spun ball	Smooth. Every sharp point is a place the coil will arc from prematurely.
Ground	Its own rod, driven outside	Never the mains earth.

Build order — and who does what

- | | |
|-----------------------------------|--|
| 1. Grounding stick | Build this <i>first</i> , before any high-voltage part exists. Hardwood or PVC handle, metal hook, heavy braid to the ground rod. Nothing else gets built until it is finished and hanging on the wall. |
| 2. Secondary | The long job and the best one to share. Wind slowly and evenly, keeping tension constant; a drill on its slowest speed with the form chucked helps. Count in hundreds and mark the form. Two coats of polyurethane when done. <i>Boys do this.</i> |
| 3. Primary | Flat spiral on an insulating cross, turns evenly spaced, with a clip lead for the tap. <i>Boys do this.</i> |
| 4. Topload | Form the duct into a ring, tape and smooth it, mount it on a plate above the secondary. <i>Boys do this.</i> |
| 5. Capacitor bank | Series-parallel string on a board, every capacitor with its bleeder resistor. <i>Boys can assemble it; adult checks every joint.</i> |
| 6. Gaps, enclosure, wiring | Main gap, safety gap, interlocked enclosure, switched and fused mains inlet, indicator lamp. <i>Adult only.</i> |
| 7. Ground rod | Driven outside, heavy strap back to the secondary base by the shortest route you can manage. |

Tuning it

1. **Calculate first.** Work out the secondary's resonant frequency from its dimensions, and set the primary tap to the turn that gives the primary tank the same f_0 . Lesson 8 is the whole of this step.
2. **Start with the gap narrow** and the tap near your calculated turn.
3. **Fire it in short bursts.** Seconds, not minutes. Unplug, short the capacitor with the grounding stick, move the tap one turn, reconnect, fire again. **Every single adjustment goes through that sequence.**
4. **Watch the streamers.** Longer and brighter means closer to tune. Work outward from the calculated turn in both directions and keep a written log of tap position against streamer length — you will not remember.
5. **Then the gap.** Once the tap is right, widen the gap a little at a time. Wider means a higher firing voltage and more energy per shot, up to the point where the transformer can no longer charge the capacitor fast enough and the output falls off again.
6. **Stop when it stops improving.** A small coil producing a clean 6–12 in streamer is a complete success and there is nothing to prove by pushing it further.

If it does not work

Symptom	Look at
No spark at the gap at all	Transformer not energized; interlock open; gap far too wide; capacitor open circuit.
Gap fires but no output	Tuning is a long way off, or the ground connection is poor or missing. Check the ground first — it is the most common fault.
Weak, thin streamers	Off tune. Move the tap. Also check for a cracked capacitor or a sooty gap.
Safety gap firing constantly	The main gap is set too wide, so voltage is reaching the safety gap first. Narrow the main gap.
Arcs from the secondary to the primary	Coupling too tight, or the topload is too small. Raise the secondary relative to the primary; add a “strike ring”.
Internal arcing in the secondary	Coupling much too tight, or turns damaged. This ruins a secondary. Stop and inspect.
Transformer getting hot	Running too long. Short bursts only.

BEFORE EVERY RUN, READ THE CHECKLIST ALOUD

Grounding stick in reach. Capacitor shorted, hands out. Enclosure closed, interlock tested. RF ground connected and tight. Everyone with an implant out of the room and all electronics elsewhere. Second person present who knows where the plug is. Safety glasses and hearing protection on. Extinguisher in reach, bench clear. Everyone beyond arm's length of the longest streamer. The named adult is the one closing the switch. The full list, and why each item is on it, is on the **Safety** sheet — and it belongs on the wall, not in a drawer.

What you built

A transformer charges a capacitor. A gap of air, switching in under a microsecond, dumps that capacitor into a tuned coil. That coil rings, and over many cycles hands its energy to a second coil tuned to the same note, loosely enough that the voltage has time to build. The result leaves the toplayer as a streamer, because the electric field at its surface has grown strong enough to tear air apart. Every clause of that is a lesson you did with a battery and a meter first — which is the only reason any of it makes sense now.

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